In [9]:

```python
# Step 1: Import Required Libraries
import pandas as pd
```

In [14]:

```python
# Step 2: Load the Dataset
df = pd.read_csv('dataset_Facebook.csv')
# Display first 5 rows
df.head()
```

Out[14]:

	pagetotallikes	type	category	postmonth	postweekday	posthour	paid	lifeti
0	139441	2	2	12	4	3	0	
1	139441	3	2	12	3	10	0	
2	139441	2	3	12	3	3	0	
3	139441	2	2	12	2	10	1	
4	139441	2	2	12	2	3	0	

In [15]:

```python
print(df.columns)
```

```
Index(['pagetotallikes', 'type', 'category', 'postmonth', 'postweekday',
       'posthour', 'paid', 'lifetimeposttotalreach',
       'lifetimeposttotalimpressions', 'lifetimeengagedusers',
       'lifetimepostconsumers', 'lifetimepostconsumptions',
       'lifetimepostimpressionsbypeoplewhohavelikedyourpage',
       'lifetimepostreachbypeoplewholikeyourpage',
       'lifetimepeoplewhohavelikedyourpageandengagedwithyourpost', 'commen
t',
       'like', 'share', 'totalinteractions'],
      dtype='str')
```

In [18]:

```python
#a. Create Data Subsets
subset_columns = df[['type', 'category', 'paid', 'like']]

print(subset_columns.head())
```

```
   type  category  paid  like
0     2         2     0    79
1     3         2     0   130
2     2         3     0    66
3     2         2     1  1572
4     2         2     0   325
```

In [19]:

```python
subset_condition = df[df['like'] > 100]

print(subset_condition.head())
```

```
   pagetotallikes  type  category  postmonth  postweekday  posthour  paid  \
1          139441     3         2         12            3        10     0
3          139441     2         2         12            2        10     1
4          139441     2         2         12            2         3     0
5          139441     3         2         12            1         9     0
6          139441     2         3         12            1         3     1

   lifetimeposttotalreach  lifetimeposttotalimpressions  lifetimeengagedus
ers  \
1                   10460                         19057                  1
457
3                   50128                         87991                  2
211
4                    7244                         13594
671
5                   10472                         20849                  1
191
6                   11692                         19479
481

   lifetimepostconsumers  lifetimepostconsumptions  \
1                   1361                      1674
3                    790                      1119
4                    410                       580
5                   1073                      1389
6                    265                       364

   lifetimepostimpressionsbypeoplewhohavelikedyourpage  \
1                                              11710
3                                              61027
4                                               6228
5                                              16034
6                                              15432

   lifetimepostreachbypeoplewholikeyourpage  \
1                                       6112
3                                      32048
4                                       3200
5                                       7852
6                                       9328

   lifetimepeoplewhohavelikedyourpageandengagedwithyourpost  comment  like
\
1                                               1108               5   130
3                                               1386              58  1572
4                                                396              19   325
5                                               1016               1   152
6                                                379               3   249

   share  totalinteractions
1     29                164
3    147               1777
4     49                393
5     33                186
6     27                279
```

In [21]:

```python
#b. Merge Data
data1 = df[['pagetotallikes', 'type', 'category']].head(10)
```

```python
data2 = df[['pagetotallikes', 'paid', 'like']].head(10)

merged_data = pd.merge(data1, data2, on='pagetotallikes')

print(merged_data.head())
```

```
   pagetotallikes  type  category  paid  like
0          139441     2         2     0    79
1          139441     2         2     0   130
2          139441     2         2     0    66
3          139441     2         2     1  1572
4          139441     2         2     0   325
```

In [22]:

```python
#c. Sort Data
sorted_data = df.sort_values(by='like', ascending=False)

print(sorted_data[['type', 'like']].head())
```

```
     type  like
252     2  1998
222     2  1639
3       2  1572
333     2  1546
101     2  1505
```

In [23]:

```python
#d. Transpose Data
transpose_data = df.head().T

print(transpose_data)
```

```
                                                          0       1       2  \
pagetotallikes                                       139441  139441  139441
type                                                      2       3       2
category                                                  2       2       3
postmonth                                                12      12      12
postweekday                                               4       3       3
posthour                                                  3      10       3
paid                                                      0       0       0
lifetimeposttotalreach                                 2752   10460    2413
lifetimeposttotalimpressions                           5091   19057    4373
lifetimeengagedusers                                    178    1457     177
lifetimepostconsumers                                   109    1361     113
lifetimepostconsumptions                                159    1674     154
lifetimepostimpressionsbypeoplewhohavelikedyour...     3078   11710    2812
lifetimepostreachbypeoplewholikeyourpage               1640    6112    1503
lifetimepeoplewhohavelikedyourpageandengagedwit...      119    1108     132
comment                                                   4       5       0
like                                                     79     130      66
share                                                    17      29      14
totalinteractions                                       100     164      80

                                                          3       4
pagetotallikes                                       139441  139441
type                                                      2       2
category                                                  2       2
postmonth                                                12      12
postweekday                                               2       2
posthour                                                 10       3
paid                                                      1       0
lifetimeposttotalreach                                50128    7244
lifetimeposttotalimpressions                          87991   13594
lifetimeengagedusers                                   2211     671
lifetimepostconsumers                                   790     410
lifetimepostconsumptions                               1119     580
lifetimepostimpressionsbypeoplewhohavelikedyour...    61027    6228
lifetimepostreachbypeoplewholikeyourpage              32048    3200
lifetimepeoplewhohavelikedyourpageandengagedwit...     1386     396
comment                                                  58      19
like                                                   1572     325
share                                                   147      49
totalinteractions                                      1777     393
```

In [24]:

```python
# e. Shape and Reshape Data
print("Shape of dataset:", df.shape)

# Reshape Data using Melt
reshape_data = pd.melt(
    df,
    id_vars=['type'],
    value_vars=['like', 'share', 'comment']
)

print(reshape_data.head())
```

```
Shape of dataset: (372, 19)
   type variable  value
0     2     like     79
1     3     like    130
2     2     like     66
3     2     like   1572
4     2     like    325
```

In [ ]: